DEEP LEARNING

What is deep learning?

Deep learning is a subset of machine learning driven by multilayered neural networks whose design is inspired by the structure of the human brain. Deep learning models power most state-of-the-art artificial intelligence (AI) today, from computer vision and generative AI to self-driving cars and robotics

How deep learning works

Deep learning uses artificial neural networks inspired by the human brain. These networks consist of many layers of connected "neurons" that process data through mathematical operations.

Key idea:

- Input data passes through multiple layers of calculations.
- Each layer transforms the data using weighted parameters.
- The model learns by adjusting these weights to reduce errors.

Important points:

- It can map inputs to outputs very effectively.
- It requires large amounts of data and computing power.
- It is often called a "black box" because its internal decisions are hard to interpret.
- Cloud computing is commonly used due to high resource needs.

Deep neural network structure

A deep neural network is made of layers of connected nodes called neurons. Each neuron applies an activation function to weighted inputs and passes the result to the next layer.

- **Layers:** Input layer → hidden layers → output layer
- **Connections:** Every neuron is linked to neurons in the next layer
- **Weights & bias:** Each connection has a weight; each neuron has a bias

- **Activation function:** Adds non-linearity so the model can learn complex patterns
- **Hidden layers:** Where most feature learning happens (simple → complex patterns)

Training deep neural networks

Training deep neural networks means improving the model by reducing prediction errors, but this is difficult because networks have millions or even billions of interconnected weights contributed to overall error is thoroughly impractical.

This obstacle was overcome by the introduction of two essential algorithms: backpropagation and gradient descent.

Backpropagation

Backpropagation, short for "backward propagation of error," is an elegant method to calculate how changes to any individual weight or bias in a neural network will affect the accuracy of model predictions.

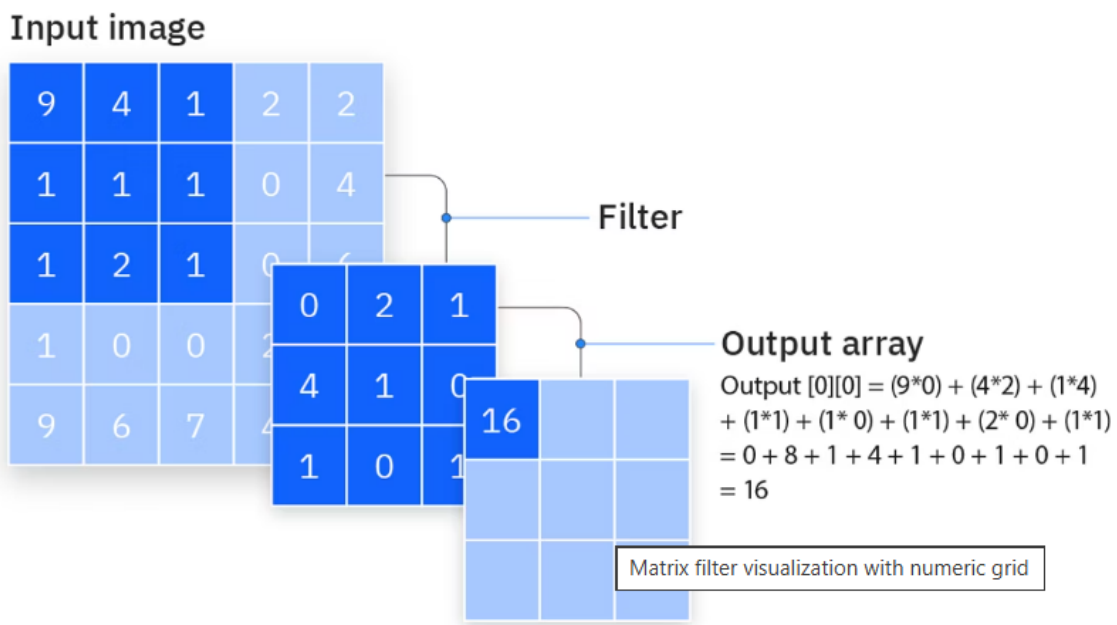
Gradient descent

The gradient computed during backpropagation then serves an input to a *gradient descent* algorithm.

Types of deep learning models

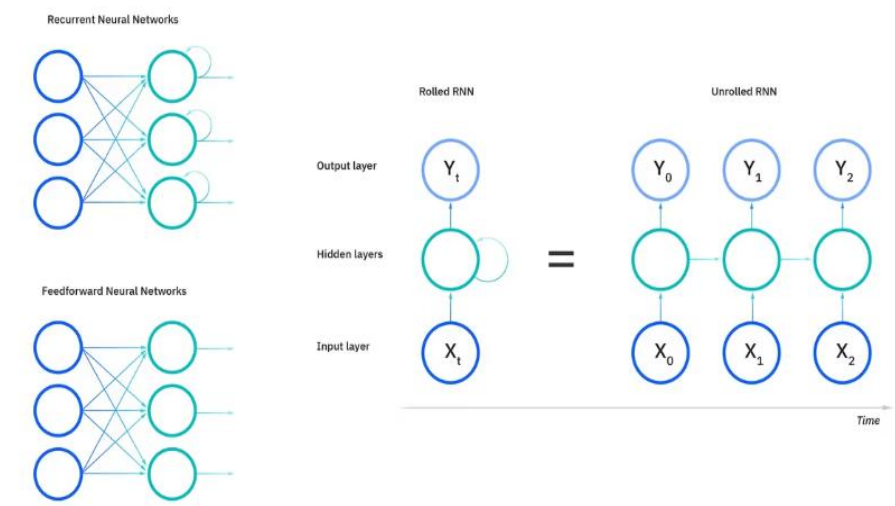
Convolutional neural networks (CNNs)

Convolutional neural networks (CNNs) are primarily (but not exclusively) associated with computer vision tasks such as object detection, image recognition, image classification and image segmentation, as they excel at "local" pattern recognition (such as relationships between adjacent pixels in an image).



Recurrent neural networks (RNNs)

Recurrent neural networks (RNNs) are used for tasks involving sequential data, such as time-series forecasting, speech recognition or natural language processing (NLP).



Used in:

• Speech recognition

• Text generation

• Time-series prediction

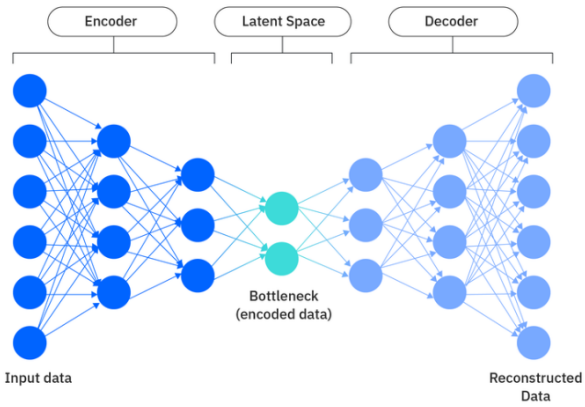
Artificial Neural Network (ANN)

An Artificial Neural Network is a model in Artificial Intelligence inspired by the human brain. It consists of interconnected neurons that process and learn from data.

- Structure: Input layer → Hidden layer(s) → Output layer
- Working: Data passes through layers, gets weighted, and produces output
- Learning: Uses Backpropagation to reduce errors
- Uses: Image recognition, speech processing, prediction tasks

Autoencoders

Autoencoders are designed to compress (or *encode*) input data, then reconstruct (decode) the original input using this compressed representation. In training, they're optimized to minimize *reconstruction loss*: the divergence between the reconstructed data point and the original input data. Though this type of deep learning uses unlabeled, unstructured data, autoencoders are generally considered to be an archetypal example of self-supervised learning.



Generative adversarial networks (GANs)

Like VAEs, generative adversarial networks (GANs) are neural networks are used to create new data resembling the original training data. GANs are a joint architecture combining two deep learning networks trained *adversarially* in a zero-sum game.